

W. Dewmini Kawshani

Software Engineer Intern

✉ dewminikawshani5@gmail.com ☎ 0775593914 📍 NO.600,Old road,Meegoda,Padukka,Sri Lanka

🌐 linkedin.com/in/dewmini-weerapperuma 🐙 github.com/dewminiweerapperuma

📝 medium.com/@dewminiweerapperuma65 🔗 dewminiweerapperuma.vercel.app

SUMMARY

IT and Management undergraduate with hands-on full-stack development experience spanning e-commerce, health-tech, and IoT projects, including AI integration. Strong problem-solver and collaborative team contributor, quick to pick up new tools and technologies, seeking a Software Engineering Internship to apply and grow these skills on real-world projects.

EDUCATION

BACHELOR OF SCIENCE HONOURS IN INFORMATION TECHNOLOGY AND MANAGEMENT 03/2024 – Present

University of Moratuwa

DIPLOMA IN ENGLISH AND LITERATURE 03/2023 – 02/2024

Aquinas Higher College of Studies

DIPLOMA IN INFORMATION TECHNOLOGY AND COMMUNICATION 03/2023 – 09/2023

ICBT campus

GCE ADVANCED LEVEL PHYSICAL SCIENCE STREAM 2023

Buddhist Girls' National College, Wennappuwa

Achieved BCC

Z-Score 0.9779

SKILLS

programming languages

JavaScript, TypeScript, Java, C, python

Frontend Development

React, Next.js, HTML5, CSS3, Responsive Web Design

Backend Development:

Node.js, Express.js, RESTful API Design, Server-Side Rendering

Databases:

PostgreSQL, MongoDB, SQL, NoSQL Data Modeling

Software Engineering Practices:

Git & GitHub, Agile/Team Collaboration, End-to-End Application Development, System Design Basics

PROJECTS

OneShop — E-Commerce & POS Platform (Team Project)

Contributed as a full-stack developer on OneShop, a combined e-commerce and point-of-sale platform sharing a synchronized MongoDB database for real-time inventory across both channels. Personally led architecture decisions including a hybrid embedding-vs-referencing schema design for shipping addresses and the choice of MongoDB over PostgreSQL to support flexible, evolving product schemas. Built core features including JWT-based authentication, role-based access control (manager/cashier), a filterable and paginated product catalog with dynamic category pages, a Sri Lanka-zone-based delivery system, and order management. Also implemented forgot-password flows via Nodemailer and led a full TypeScript conversion of key product components.

(Next.js 14, TypeScript, Node.js/Express, MongoDB Atlas, JWT, Nodemailer)

LottoScan — Sri Lankan Lottery Ticket Checking Platform (Ongoing)

Developing a full-stack web application for checking Sri Lankan lottery results (NLB/DLB), supporting all 16 national lottery types with complete prize structures. Implements QR code scanning and manual number entry for ticket verification, automated result scraping synced to official draw times, and Sinhala-to-romanized-English name conversion for localized display. Built with Next.js and Express.js, using QuestDB for time-series result data and Docker Compose for containerized orchestration; includes PWA support and automated winner email notifications.

(Next.js, Express.js, QuestDB, Docker Compose, PWA, QR scanning, web scraping automation)

Small Shop Inventory + Reorder Alert System

Built a full-stack inventory management platform for small shop owners to replace manual stock tracking, featuring barcode scanning via phone camera for scan-to-sell workflows, automatic stock decrementing, and low-stock reorder alerts. Designed a relational schema with PostgreSQL (Supabase) and Prisma ORM for data integrity across stock and sales records, with forward-looking support for stock movement logs, supplier price history, and batch/expiry tracking. Implemented role-based access control for owner/staff permissions and a sales dashboard surfacing best- and worst-selling items.

(Next.js 14, TypeScript, Node.js/Express, PostgreSQL, Prisma, Supabase — barcode scanning, role-based access control, relational schema design)

🔗 — AI-Powered Menstrual Health & Cycle Tracking Application (Two-Person Team Project)

(ongoing)

Developing RESTful backend services using Node.js and Express.js for cycle tracking, symptom logging, user profiles, and personalized insights. Designing the Supabase PostgreSQL database with relational schemas, Row-Level Security (RLS) policies, validation, and secure user-level data access. Building a cycle prediction algorithm supporting regular and irregular cycles, incorporating average cycle length, variability, standard deviation, predicted ranges, and confidence levels. Integrating the Claude API to build a context-aware, non-diagnostic chatbot with personalized responses and red-flag symptom detection, while collaborating on API contracts, frontend integration, and cloud deployment.

(Node.js, Express.js, JavaScript, PostgreSQL/Supabase, Claude API, REST APIs, Row-Level Security, Git/GitHub, Cloud Deployment)

Indoor Air Quality Monitoring System (Team Project)

Built a real-time indoor air quality monitoring system on ESP32-WROOM-32, measuring particulate matter (PM1.0/2.5/10), VOCs/NOx, O3, CO2, temperature, and humidity via a multi-sensor array (PMS5003, SGP41, ZMOD4510, MH-Z19, DHT22, DS18B20). Calculated real-time AQI based on Sri Lankan air quality standards, with local flash storage and RTC-timestamped logging to handle connectivity outages, and dual Wi-Fi/Ethernet redundancy for reliable cloud sync to Firebase. Developed a React + Vite web dashboard for real-time and historical data visualization, alongside a custom PCB (EasyEDA) and 3D-modeled enclosure (Blender) for a compact, user-friendly device.

(ESP32, C/C++ (Arduino IDE), Firebase Realtime Database, React, Vite, EasyEDA, Blender)

COURSES

web design for beginners

Centre for open and distance learning, University of Moratuwa

Python for Beginners

Centre for open and distance learning, University of Moratuwa

(progressing)

Data Science for beginners

IBM

(progressing)